

Case 2

A pit in Queensland was considered for highwall mining. It was primarily considered for use of a CHM system due to the availability of a system on site.

A summary of the general information regarding the pit is given below:

- ❑ roof condition was expected to be poor, with the immediate roof being weak, closely laminated mudstone;
- ❑ mineable reserve was 900 m (long) × 300 m (penetration);
- ❑ seam gas level was low;
- ❑ seam dip was about 4-5°;
- ❑ pit highwall was generally smooth, no risk of major highwall/lowwall instability;
- ❑ seam continuity was expected to be good but with low confidence level;
- ❑ pit condition was fair;
- ❑ coal quality was excellent;
- ❑ floor was soft mudstone and may cause some problems for highwall mining;
- ❑ it was not known whether ground water may be a problem;
- ❑ coal strength was low (UCS = 10 MPa);
- ❑ cover depth was in the range of 50-80 m;
- ❑ there was no major environmental concern for subsidence;
- ❑ seam thickness was around 2 m
- ❑ mining equipment and crews were available on site.

The evaluation of HMI is given in **Error! Reference source not found.** for CHM. An HMI of 50 and a confidence level of 61% are obtained. The results put this pit in the Transition Zone, see **Error! Reference source not found.**

Based on the HMI evaluation, this pit would not be considered very attractive for highwall mining using CHM. The potential problems identified are the poor roof and floor conditions and low coal strengths. However, the pit need not be totally rejected for highwall mining using CHM.

The mine decided to mine this pit using CHM mainly due to non-technical reasons (namely. the availability of the CHM system on site and a desire to continue CHM operations at the mine). Only limited success was achieved. Frequent roof falls caused early termination of many entries. Also as a result of the interaction between roof falls and pillar failure, severe roof collapses occurred in several entries during mining which led to the loss of mining equipment. In another part of the mining area, a panel failure occurred after the completion of mining.

Case 3

An open cut mine in Queensland considered using a CHM to mine coal. A pit with mineable highwall length of 530 m was identified as the first possible site. There was very limited geological and geotechnical data available due to the relatively small mining scale and short mining history. There was no previous highwall mining experience at the mine. The following information was based on limited drill hole data and observations of the pit:

- ❑ immediate roof was a 0.8 m weak to moderately strong claystone, closely bedded. Roof jointing condition was unknown;
- ❑ mineable reserve was 530 m along highwall but was confined by a regional fault to a variable penetration between 50 m and 300 m;
- ❑ seam gas level may be medium to high but it was uncertain;
- ❑ seam dip was expected to be in the range 5-10° down dip;
- ❑ pit highwall was generally smooth, no risk of major highwall/lowwall instability;
- ❑ seam continuity was expected to be reasonable, but could be affected by possible structures formed together with the regional fault;
- ❑ pit condition was good;
- ❑ seam quality was good in the upper part but high ash in the lower part;
- ❑ floor was a layer of intact mudstone, underlain by siltstone;
- ❑ it was not known whether ground water would be a problem;
- ❑ coal strength could be low but there was no laboratory test data available. Coal joints were extensive and closely spaced;
- ❑ cover depth was in the range of 50-80 m;
- ❑ there was no major environmental concern about subsidence except for the possibility of draining a small local reservoir in the mining reserve;;
- ❑ seam thickness was around 4 m;
- ❑ mining equipment and crews were available in nearby mines.

The evaluation of HMI is given in **Error! Reference source not found.**. An HMI of 64 and a confidence level of 39% are obtained. The results put this pit in Region 2, see **Error! Reference source not found.**. This pit is likely to be a suitable site for CHM mining. The potential problems identified could be the poor roof and floor conditions, uncertain seam gas and low coal strengths. The lack of information resulted in a low confidence level of the HMI. To confirm the initial evaluation results, more site investigation was needed to better evaluate the seam conditions (thickness, dip, quality, strength gas etc.).

The pit was mined after a limited site investigation that was constrained by budget. The mining operation was not very successful. The average penetration achieved was only about 150 m, far less than the designed 300 m. A number of problems contributed to the short penetration, including gas, roof falls, unexpected geological structures, and possible pillar failures resulting in highwall cracking. A localised subsidence occurred one year after mining and it drained half of the small reservoir on top of the mined areas.